

# 02456 Deep Learning

## Exercises Week 4

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**Prince 10.1** Show that the operation in equation 10.4 is equivariant wrt translation;

Equation 10.4 defines the convolution operation for a single hidden unit  $h_i$  as

$$h_i = a \left[ \beta + \sum_{j=1}^3 \omega_j x_{i+j-2} \right]$$

Equation 10.2 defines equivariance as  $f[t[x]] = t[f[x]]$ , where  $f$  is our function and  $t$  is the translation (i.e. shifting the input). To show that equation 10.4 is equivariant wrt translation is thereby proving equation 10.2 for  $f = \text{equation 10.4}$ . We thereby define a sequence as  $x$  shifted by some constant  $c$  as  $\tilde{x}_i = x_{i-c}$ . This means that the element at position  $i$  in the new sequence corresponds to the element at position  $i - c$  in the original sequence. If we use this as input in equation 10.4, we get

$$\tilde{h}_i = a \left[ \beta + \sum_{j=1}^3 \omega_j \tilde{x}_{i+j-2} \right]$$

This corresponds to  $f[t[x]]$  since we apply the translation first, and then the convolution function. If we instead apply the convolution function first we get  $h_i$  and then we have to shift that by  $c$  getting us  $h_{i-c}$ . Since  $\tilde{h}_i = h_{i-c}$  we have proven that it does not matter if you apply the translation before or after the convolution, the result is identical.

**Prince 10.6** Draw a  $6 \times 12$  weight matrix in the style of figure 10.4d relating the inputs  $x_1, \dots, x_6$  to the outputs  $h_1, \dots, h_{12}$  in the multi-channel convolution as depicted in figures 10.5a-b;

$$\begin{bmatrix} \omega_2 & \omega_3 & 0 & 0 & 0 & 0 \\ \omega_1 & \omega_2 & \omega_3 & 0 & 0 & 0 \\ 0 & \omega_1 & \omega_2 & \omega_3 & 0 & 0 \\ 0 & 0 & \omega_1 & \omega_2 & \omega_3 & 0 \\ 0 & 0 & 0 & \omega_1 & \omega_2 & \omega_3 \\ 0 & 0 & 0 & 0 & \omega_1 & \omega_2 \\ \omega_5 & \omega_6 & 0 & 0 & 0 & 0 \\ \omega_4 & \omega_5 & \omega_6 & 0 & 0 & 0 \\ 0 & \omega_4 & \omega_5 & \omega_6 & 0 & 0 \\ 0 & 0 & \omega_4 & \omega_5 & \omega_6 & 0 \\ 0 & 0 & 0 & \omega_4 & \omega_5 & \omega_6 \\ 0 & 0 & 0 & 0 & \omega_4 & \omega_5 \end{bmatrix}$$

**Prince 10.16** Consider the AlexNet network (figure 10.16). How many parameters are used in each convolutional and fully connected layer? What is the total number of parameters?;

The total number of parameters for a convolutional layer is computed as

$$(\text{kernel height} \times \text{kernel width} \times \text{input channels} \times \text{output channels}) + \text{output channels}$$

The first layer in the AlexNet from figure 10.16 uses a  $11 \times 11$  kernel. The input is a  $224 \times 224$  image in 3 channels, and layer produces 96 output channels. The first layer has  $11 \cdot 11 \cdot 3 \cdot 96 + 96 = 34,944$  parameters. Continuing in the same pattern, the following layers have 614,656, 885,120, 1,327,488, and 884,992 parameters. The output from the last convolutional layer is resized into a single vector with length 43,264, such that the parameter calculations for the fully connected layers FC1, FC2, and FC3 is; (1)  $43,264 \cdot 4,096 = 177,213,440$  (2)  $4,096 \cdot 4,096 + 4,096 = 16,781,312$  (3)  $4,096 \cdot 1,000 + 1,000 = 4,097,000$ . The total parameters for AlexNet is then calculated as the sum of all these, 201,838,952. *Note:* the book claims the complete network contains  $\sim 60$  million parameters. This is probably because the MaxPool before the FC's reduces the input such that the vector is not really 43,264 as stated.

**Exam 2025 Q5** Consider a 2D convolutional layer with no padding, a stride of 2 (both height and width), no bias, and the kernel

$$K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

The input to the layers is

$$X = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 2 & 2 \\ 0 & 2 & 1 & 1 \end{bmatrix}$$

What is the output of the layer?

- A.  $\begin{bmatrix} 3 & 2 \\ 0 & 3 \end{bmatrix}$
- B.  $\begin{bmatrix} 4 & 2 \\ 3 & 6 \end{bmatrix}$
- C.  $\begin{bmatrix} 1 & 0 \\ 3 & 3 \end{bmatrix}$
- D.  $\begin{bmatrix} 1 & 3 & 0 \\ 1 & 2 & 3 \\ 3 & 1 & 3 \end{bmatrix}$

C is the correct. A stride of 2 in both height and width (or horizontally and vertically) means we move the filter two right a time, and 2 down when we go down. Thus, the resulting output is a  $2 \times 2$  matrix.

**Exam 2025 Q6** Consider a 2D convolutional layer with padding 1 (height and width), a stride of 2 (height and width), no bias, and 10 kernels of size (`channels_in`, `kernel_height`, `kernel_width`) = (5, 3, 3). For an input tensor of size (`channels`, `height`, `width`) = (5, 5, 5), what is the output size of the layer?

- A. (`channels`, `height`, `width`) = (10, 2, 2)
- B. (`channels`, `height`, `width`) = (10, 3, 3)
- C. (`channels`, `height`, `width`) = (10, 5, 5)
- D. (`channels`, `height`, `width`) = (5, 3, 3)

B is correct. Since the convolutional layer has 10 kernels, its output is across 10 channels. Now, the input tensor is a  $5 \times 5$  matrix with a padding of 1, effectively making it a  $7 \times 7$  matrix. And since the  $3 \times 3$  kernel moves with a stride of 2 both vertically and horizontally, it will sample 3 times in both dimensions of the input giving a  $3 \times 3$  output.